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RESEARCH FOCUSES

- Biocompatible materials for human health
- Neuromodulation, neurostimulation, nerve conduction block
- Wireless energy transfer for implantable medical devices
- Pathogen control with static electricity
- Energy harvesting for sensor applications

DEGREE & EDUCATION

Postdoctoral Fellow in Medical Engineering **Sep 2025 –**

California Institute of Technology (Caltech), Advisor: Prof. Wei Gao

Postdoctoral Fellow in Advanced Functional Polymers Research Center **Sep 2023 – Jul 2025**

Korea Research Institute of Chemical Technology (KRICT), Advisor: Dr. Yun Ho Kim

Ph.D in Advanced Material Science & Engineering **Mar 2019 – Aug 2023**

Sungkyunkwan University (SKKU), Advisor: Prof. Sang-Woo Kim

* Ph.D. thesis title: Design and Development of Triboelectric Nanogenerator for Pathogen Control and Powering Implantable Device

B.S. in Mechanical Engineering **Mar 2013 – Feb 2019**

Hanyang University (HYU), Advisor: Prof. Se-Jin Yook

AWARD & HONORS

- **Sejong Science Fellowship** (Domestic Track, 2024-2029, **100,000,000 KRW/year**)
- PSK-INNOX Young Researcher Award by the Korea Polymer Society of Korea (2024)
- Samsung Humantech Silver Medal (2023)
- The Best Outstanding Master/Ph.D Student Award, BK21 Program (2022)
- Best Presentation Award (NCC 2022)
- Excellent Student Presentation Award (ECS 242nd Meeting, 2022)
- SKKU Research Matters (2021)
- SKKU Best Paper Award (2021)
- Best Paper Award (PVSEC-30 & GPVC 2020)
- Energy Technology Innovation Ideas (Ministry of Commerce Industry and Energy, 2019)
- Outstanding Poster Presentation Award (12th TBIS, 2019)

PUBLICATIONS

PEER-REVIEWED JOURNALS

Notes: # indicates equally contributing authors. * indicates the corresponding author(s).

First Authored Papers

1. **Y.-J. Kim#**, Y. Chung#, J.-H. Hwang#, J.-M. Jeong, B. Kim, D. Kang, B.-O. Choi, H.-J. Yoon*, S.-W. Kim*, Acoustic Impedance-Tailored High-Performance Ultrasound-Driven Triboelectric Nanogenerators, ***Advanced Energy Materials***, Early view (2026)

2. **Y.-J. Kim#**, S.-H. Kim#, B.-J. Park#, J. Jeon, D. Kang, Y. Chung, J.-H. Hwang, H.-J. Yoon, K. H. Lee*, B.-O. Choi*, S.-W. Kim*, Wireless and bioresorbable triboelectric nerve block system for postoperative pain control, ***Nature Biomedical Engineering*** (2026), *Covered by ScienMag, Bioengineer, and domestic news*

3. **Y.-J. Kim#**, J. Hwang#, B.-J. Park, Y. Chung, S. H. Choi, S.-H. Kim, D. H. Park, Y. Lee, S. Yoo, J. Park, J. C. Won, S. H. Nam, B.-O. Choi, S.-W. Kim*, Y. H. Kim*, Nanoparticle-Stabilized Porous Composite Architectures for Ultrasound-Driven Triboelectric Energy Harvesters in Soft Implantable Electronics, ***ACS Nano*** 19, 40994–41003 (2025)

4. N. G. Kim#, J. M. Jang#, **Y.-J. Kim#**, J. Y. Kim, D. Kang, J. Park, J. C. Won, S.-W. Kim, D. W. Kim, S. J. Kim, Y. H. Kim, Scalable and Eco-friendly Production of Flexible, Hierarchical Porous MXene/Polyimide Composites with Superior Electromagnetic Interference Shielding, ***Composites Part B: Engineering*** 304, 112805 (2025)

5. Y. Chung#, H. Yuan#, Z. Wang, J.-M. Jeong, B.-J. Park, J.-H. Hwang, S.-J. Suh, B.-O. Choi, H. Park, **Y.-J. Kim***, K. Dai*, S.-W. Kim*, Acoustic Tunable Battery-Free Implants Based on Sustainable Triboelectric Nanogenerators with Metal-Polymer Intermixing Layers, ***Advanced Energy Materials*** 15, 2403712 (2025), *selected as a **cover image** of the journal issue*

6. **Y.-J. Kim#**, Z.-Y. Huo#*, H. Dai#, D.-M. Lee, I.-Y. Suh, J.-H. Hwang, Y. Chung, H. Y. Lee, Y. Du, W. Ding, X. Wang, S.-W. Kim*, Body-coupled energy enabling unrestricted microbial disinfection using polymer nanorods, ***Nature Water*** 2, 360-369 (2024), *Covered by News & Views in **Nature Water**, Highlighted in **Nature*** 628, 693 (2024)

7. **Y.-J. Kim#**, J. Lee#, J.-H Hwang#, Y. Chung, B. J. Park, J. Kim, S.-H. Kim, H.-J. Yoon, S.-M. Park*, S.-W. Kim*, High-performing and capacitive-matched triboelectric implants driven by ultrasound, ***Advanced Materials* 36**, 2307194 (2024), *Covered by **Forbes**, Parkinson's News Today, News-medical, etc.*

8. I.-Y. Suh#, **Y.-J. Kim#**, B.-Q. Chen, P. Zhao, D. S. Cho, M. Kang, Z.-Y. Huo*, S.-W. Kim*, Self-powered microbial blocking textile driven by triboelectric charges, ***Nano Energy*** 110, 108343 (2023)

9. **Y.-J. Kim#**, H. T. Kim#, J. H. Lee#, I.-Y. Suh, S.-W. Kim*, Self-powered Fine Dust Filtration System Using Triboelectrification Induced Electric Field, ***Nanoscale Research Letters*** 17, 1-9 (2022)

10. M. Kang#, N.-Y. Jang#, **Y.-J. Kim#**, H.-J. Ro#, D. Kim, Y. Kim, H. T. Kim, H. M. Kwon, J.-H. Ahn, B.-O. Choi, N.-H. Cho*, and S.-W. Kim*, Virus blocking textile for SARS-CoV-2 using human body triboelectric energy harvesting, ***Cell Reports Physical Science*** 3, 100813 (2022)

11. Z.-Y. Huo#, **Y.-J. Kim#**, I.-Y. Suh, D.-M. Lee, J. H. Lee, Y. Du, S. Wang, H.-J. Yoon, and S.-W. Kim*, Triboelectrification induced self-powered microbial disinfection using nanowire-enhanced localized electric field, ***Nature Communications*** 12, 1-11 (2021) *Covered by domestic news (Dong-A Ilbo, etc.)*

Contributing Authored Papers

12. Y. Lin, J. Zhou, I.-Y. Suh, E. Kim, K. C. Yip, D. Kim, J.-M. Jeong, **Y.-J. Kim**, J. Jeon, J.-H. Lee, Y.-H. Hwang, S.-W. Kim, D.-M. Shin, Concurrently Achieving 4.6 W/M^2 and 120,000 Cyclability Enabled by Extendable Swing Arms in Rotational Triboelectric Nanogenerator, ***Advanced Science***, Early view (2026)

13. Y. Lee, **Y.-J. Kim**, D. H. Park, I. Park, D. Kim, S. B. Kim, Compact electroaerodynamic dust removal through integrated ion transport mechanisms, ***Separation and Purification Technology*** 386, 136559 (2026)

14. I.-Y. Suh, **Y.-J. Kim**, S.-W. Kim*, Portable hand-powered nanocatalysis for clean water, ***Nature Nanotechnology*** 21, 15-16 (2025)

15. Y. Chung, S. Choudhury, **Y.-J. Kim**, I.-Y. Suh, K. Dai, S.-W. Kim, Ultrasound-Powered, Battery-Free Implants via Triboelectric Energy Harvesting, ***Advanced Materials Technologies*** early view, e01451 (2025)

16. Z.-Y. Huo, **Y.-J. Kim**, S.-W. Kim, Contact electrification-induced personal sanitation, ***Nature Reviews Clean Technology*** 1, 673-674 (2025)

17. D. Kang#, B.-J. Park#, J.H. Hwang#, **Y.-J. Kim**, S.-H. Kim, H. W. Kim, K. J. Yu, J. Jeon, H. Y. Lee, Y. Chung, S. H. Nam, B.-O. Choi, S.-W. Kim, In-body current path manipulation with minimal attenuation, ***Science Advances*** 11, eadx5922 (2025)

18. D. Kim#, Y. Kwon#, Y. So, **Y.-J. Kim**, S. W. Park, H. Park, J. Hwang, J. Park, C. Kim, J. C. Won, J. H. Kim*, Y. H. Kim*, Water-Borne Fluorinated Polyimide Dielectric for Large-Area IGZO Transistors and Logic Gates, ***ACS Applied Materials & Interfaces*** 16, 68328-68335 (2024)

19. Y. Chung#, J.-M. Jeong#, J.-H. Hwang#, **Y.-J. Kim**, B.-J. Park, D. S. Cho, Y. Cho, S. J. Suh, B.-O. Choi, H. Park*, H.-J. Yoon*, S.-W. Kim*, Gigantic triboelectric power generation overcoming acoustic energy barrier using metal-liquid coupling, ***Joule*** 8, 1-15 (2024)

20. M. Kang#, H. Yum#, H. T. Kim#, B. J. Park, Y. Choi, H. J. Kim, Y. Cho, **Y.-J. Kim**, D.-M. Lee, D.-G. Lee, H.-C. Song, S. H. Nam, J. H. Lee*, B.-O. Choi*, S.-W. Kim*, Self-Powered Electrical Bandage Based on Body-Coupled Energy Harvesting, ***Advanced Materials*** 36, 2402491 (2024)

21. I.-Y. Suh#, Z.-Y. Huo#*, J.-H. Jung#, D. Kang, D.-M. Lee, Y.-J. Kim, B. Kim, J. Jeon, P. Zhao, J. Shin, S. Kim, S.-W. Kim*, Highly efficient microbial inactivation enabled by tunneling charges injected through two-dimensional electronics, ***Science Advances*** 10, eadl5067 (2024)

22. J. Kim#, D.-M. Lee#, H. Ryu, **Y.-J. Kim**, H. Kim, H.-J. Yoon, M. Kang, S. S. Kwak*, S.-W. Kim*, Triboelectric Nanogenerators for Battery-Free Wireless Sensor System Using Multi-Degree of Freedom Vibration, ***Advanced Materials Technologies*** 9, 2301427 (2024)

23. D. Kang#, J.-H. Hwang#, **Y-J. Kim**, Z. Pin, H. Y. Lee, J. Kim, M. S. Shin, S. Jeon, S. Kim, S.-W. Kim*, Contact electrification controlled by material deformation-induced electronic structure changes, ***Materials Today*** 72, 109-116 (2023)

24. B. Kim#, H.-J. Yoon#, **Y.-J. Kim**, B.-J. Park, J.-H. Jung, S.-W. Kim*, Ultrasound-Driven Triboelectric Nanogenerator with Biocompatible 2-Hydroxyethyl Methacrylate, ***ACS Energy Letters*** 8, 3412 (2023)

25. Z.-Y. Huo, **Y.-J. Kim**, Y. Chen, Y. Du, D. Xia, S.-W. Kim*, Q. Yuan*, Hybrid energy harvesting systems for self-powered sustainable water purification by harnessing ambient energy, ***Frontiers of Environmental Science & Engineering*** 17, 118 (2023) (*review articles*)

26. X. Meng#, X. Xiao#, S. Jeon, D. Kim, B.-J. Park, **Y.-J. Kim**, N. Rubab, S. Kim, S.-W. Kim*, An Ultrasound-Driven Bioadhesive Triboelectric Nanogenerator for Instant Wound Sealing and Electrically Accelerated Healing in Emergencies, ***Advanced Materials*** 35, 2209054 (2023)

27. I. M. Imani#, B. Kim#, X. Xiao, N. Rubab, B.-J. Park, **Y.-J. Kim**, P. Zhao, M. Kang, S.-W. Kim*, Ultrasound-Driven On-Demand Transient Triboelectric Nanogenerator for Subcutaneous Antibacterial Activity, ***Advanced Science*** 10, 2204801 (2023)

28. B.-G. Park#, C. Lee#, **Y.-J. Kim**, J. Park, H. Kim, Y. Jung, J. S. Ko, S.-W. Kim, J.-H. Lee*, H. Cho*, Toxic micro/nano particles removal in water via triboelectric nanogenerator, ***Nano Energy*** 100, 107433 (2022)

29. M. Kang#, M. S. B. Khusrin#, **Y.-J. Kim**, B. Kim, B.-J. Park, I. Hyun, I. M. Imani, B.-O. Choi, S.-W. Kim*, Nature-derived highly tribopositive κ-carrageenan-agar composite-based fully biodegradable triboelectric nanogenerators, ***Nano Energy*** 100, 107480 (2022)

30. Z.-Y. Huo#, D.-M. Lee#, J.-M. Jeong#, **Y.-J. Kim**, J. Kim, I.-Y. Suh, P. Xiong, S.-W Kim*, Microbial Disinfection with Supercoiling Capacitive Triboelectric Nanogenerator, ***Advanced Energy Materials*** 12, 2103680 (2022)

31. D.-M. Lee#, N. Rubab#, I. Hyun, W. Kang, **Y.-J. Kim**, M. Kang, B. O. Choi*, S.-W. Kim*, Ultrasound-

mediated triboelectric nanogenerator for powering on-demand transient electronics, ***Science Advances*** 8, eabl8423 (2022)

32. Z.-Y. Huo, D.-M. Lee, **Y.-J. Kim**, and S.-W. Kim*, Solar-induced hybrid energy harvesters for advanced oxidation water treatment, ***iScience*** 24, 102808 (2021) (*review articles*)

33. Z.-Y. Huo, D.-M. Lee, S. Wang, **Y.-J. Kim**, and S.-W. Kim*, Emerging Energy Harvesting Materials and Devices for Self-Powered Water Disinfection, ***Small Methods*** 5, 2100093 (2021) (*review articles*)

34. H.-J. Yoon#, D.-M. Lee#, **Y.-J. Kim**, S. Jeon, J.-H. Jung, S. S. Kwak, J. Kim, S. Kim, Y. Kim, and S.-W. Kim*, Mechanoreceptor-Inspired Dynamic Mechanical Stimuli Perception based on Switchable Ionic Polarization, ***Advanced Functional Materials*** 31, 2100649 (2021)

PATENTED TECHNOLOGY TRANSFER

1. KR Application No. 10-2020-0062731, No. 10-2020-0140031, **SEMS Co., Ltd., (85,000,000 KRW)**
2. KR Application No. 10-2020-0099247, **SEMS Co., Ltd., (58,000,000 KRW)**
3. KR Application No. 10-2020-0056913, US Application No. 17/110,673, **SEMS Co., Ltd., (79,000,000 KRW)**
4. KR Application No. 10-2020-0145170, US Application No. 17/510,675 **Energymining Co., Ltd., (300,000,000 KRW)**
5. KR Application No. 10-2024-0137352, No. 10-2023-0159093, **Edge Foundry Co. Ltd., (55,000,000 KRW)**

PATENTS:

	Application (KOREA)	Registration (KOREA)	Application (US)	Registration (US)
Patent (#)	6	24	7	5

REFERENCES:

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